Adicherla Shiva Varun

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.20
ALPHORES JUNIOR COLLEGE	Class XII (Telangana State Board of Intermediate Education)	2024	93.60%
SIDDARTHA HIGH SCHOOL	Class X (Telangana State Board of Secondary Education)	2022	10.00%

PROJECTS

FPGA Based Maze Solving Bot

- Architected digital logic in Verilog to develop a differential drive robot capable of solving a maze.
- Written custom Verilog logic for rotary encoder, and PWM motor control.
- Implemented an FSM combining ultrasonic sensor and rotary encoder data, for localisation and realtime wall avoidance.
- Debugged and patched brownout issues due to the motor driver by using safe FSM encoding.
- Designed and fabricated the mechanical frame for the bot.

Low Power BLE Beacon

- Developed a low-power air quality monitoring beacon achieving ~3 days of battery life on a 1000mAh Li-po cell.
- Integrated three I2C air quality sensors (SCD40, SGP41, BMV080) with custom drivers.
- Used NimBLE for BTHome V2 protocol extensions to transmit data to mobile applications and Home Assistant Nodes.
- Optimized data encoding using bit-packing to fit readings within the 31-byte BLE legacy advertising packet limit.
- Designed custom PCB layout in Altium, optimizing power distribution and small footprint considering enclosure.
- Debugged deep sleep wake cycles, managing RTC memory persistence for sensor calibration data across power states.
- Designed and 3D printed a custom enclosure meeting the integration guidelines of gas sensors.

Auto-Ranging LCR Meter

- Designed a LCR meter capable of measuring Inductance, Capacitance, and Resistance within a <10% error margin.
- Synthesized an AC excitation source using ESP32's internal DAC and an op-amp low-pass filter and unity-gain buffer.
- Implemented auto-ranging gain stages via analog multiplexers to increase measurement range.
- Applied IQ demodulation in firmware to reliably extract magnitude and phase shift even with noisy ADC.

1D FEM Schrödinger-Poisson solver

- Implemented a 1D Mesh generator, Finite Element Method for solving Poisson and Schrödinger's equation.
- Wrote a lightweight custom matrix library accelerated with OpenMP for solving sparse assembled equations.
- Tested and validated the solver for analytic correctness using existing solvers.

SKILLS

Tools: Linux, Vim, Git, GNU Make, Podman/Docker, Quartus Prime, OnShape (MCAD)

Programming languages: C/C++, Python, Verilog

Simulations: LTSpice/NgSpice, Icarus Verilog

PCB Design: Altium, KiCAD

Documentation: LaTeX, Typst

RELEVANT COURSES

Theory: Circuits and Network Analysis, Microprocessor and Computer Architecture, Signals and Systems, Digital Systems,

Probability and Random Processes, Matrix Theory, Semiconductor Device Fundamentals, Numerical Methods for Device

Modelling, Modelling of Nanoscale Transistors, Productivity Tools for IC design, Analog Electronics, Introduction to CMOS

Processing,Control Theory, Engineering Electromagnetics, Introduction to Programming.

Lab: Digital Systems Lab, Semiconductor Devices Lab, Electronic Devices and Circuits Lab, Analog Circuits Lab, Electronic System Design Project Lab, PCB Design and Layout for electronic systems.

POSITIONS OF RESPONSIBILITY

UG Mentor - Sunshine IITH

- Conducted engaging meets and games aimed at 27 freshmen to help ease transition to their college life.
- Provided guidance on academic planning, course management, and their academic projects.

ACHIEVEMENTS

MELCO Hackathon - IITH

- Scanned and generated synthetic point cloud data to train a neural network achieving an IoU of ~50%, winning 2 of 6 categories and receiving an award of ¥100,000.